

Navya

Honest benchmark numbers

RTX 4060

Stage 2 PoC - 1-2 Sept

SIH26104 - Team Sonix

Your mission: Produce the two generalisation numbers we quote to the jury - and protect their honesty.

THE SITUATION - READ THIS FIRST

Where we are. SONIX listens to a call in 4-second chunks and shows green / amber / red. On the academic benchmark it is strong: **1.49% EER** on ASVspoof-2019 LA. But on our own real recordings it wrongly flags roughly **half of genuine voices as fake**.

Why. The model only ever learned from clean studio recordings of real people - same good microphone, same quiet room. Our phone and laptop recordings sound nothing like that. So the model thinks: "this is not like any real voice I know, therefore fake." **The model is not broken. It is narrow.** It has never heard normal life.

The fix - three things. (1) **Show it more kinds of real:** add genuine recordings from many phones, mics, rooms and languages to training. (2) **Make it ignore silence:** quiet gaps are what break the "minimal voice" clips. (3) **Reset the alarm levels:** our amber/red numbers came from the clean test, so we re-set them using real recordings.

HARD RULE: we never train on In-the-Wild or on any eval set. Those are our honest test. Training on them would destroy the one number that proves we generalise.

THE PLAN - THREE LANES RUNNING AT ONCE

LANE 1 - FIX THE MODEL

Silence gate, more real audio, RawBoost augmentation, retrain, recalibrate. This is what stops the false alarms.

Yugal - Akshat - Yukti

LANE 2 - HONEST NUMBERS

2021-DF EER, In-the-Wild EER, codec 2x2, metrics + calibration. These are the numbers we quote to the jury.

Navya - Anubhav - Yukti

LANE 3 - A DEMO THAT CANNOT FAIL

Replay mode on validated clips, clip library, pitch, Q&A, backup video. The demo never depends on live luck.

Suryansh - Anubhav - Akshat

YOUR TASKS

P1 NOW

Get the ASVspoof-2021 DF CM key

- This is the project's one hard blocker. Without the key file that maps trial IDs to bonafide/spoof, there is NO EER - only meaningless raw scores.
- If it is not already downloaded, tell Yugal immediately, before doing anything else.

Done when: CM key file present on disk.

P1 TONIGHT

Consolidate DF embeddings from the SSD and score

- Copy the 4 embedded parts into outputs\embeddings_df21\eval.
- Check every shard_*.npz has its matching .labels.npz, or that trial IDs survived so labels can be joined.
- Score with baseline and augmented heads -> DF21 EER.

```
outputs\embeddings_df21\eval\shard_*.npz (+ .labels.npz)
python src\eval.py --split df21 --model-ckpt outputs\models\head.pt
```

Done when: A DF21 EER number for baseline and augmented.

P1 TONIGHT

In-the-Wild EER - TEST ONLY

- The itw embeddings already exist at outputs\embeddings\itw. Join labels from the ITW meta file and score.
- READ THIS TWICE: we NEVER train on In-the-Wild. It is our honest generalisation test. If we train on it, the number becomes worthless and the jury is right to call it out.
- Expect this number to be much worse than 1.49%. That is normal and expected for every model of this class - we report it honestly.

```
python src\eval.py --split itw --model-ckpt outputs\models\head.pt
```

Done when: An honest ITW EER, reported as-is.

P2 TOMORROW

Take a RawBoost extraction batch

- Once your scoring is done your 4060 is free. Yugal will send you a slice of the RawBoost audio to extract embeddings for.

```
python src\extract_embeddings.py --split train --data-root data\asvspoof19_la_rawboost --out outputs\embeddings_rawboost --batch 8
```

Done when: Your slice of rawboost embeddings delivered.

YOUR CHECKLIST

- ☐ DF21 CM key obtained
- ☐ DF21 EER (baseline + augmented)
- ☐ ITW EER, test-only, untouched by training
- ☐ RawBoost extraction slice done

Note: Raise the CM key blocker LOUDLY and early if it is missing.

WORDS WE KEEP USING

EER	One number for how wrong the model is. Lower = better. 1.49% is strong. 50% = coin flip.
bonafide	A real human voice.
spoof	An AI-cloned / synthetic voice.
embedding	The 1024-number fingerprint of a 4-second audio chunk.
head	The small model we actually train. The big wav2vec2 front-end is frozen (never trained).
VAD	Voice Activity Detection - is anyone actually speaking in this chunk?
augmentation	Making altered copies of training audio (phone-compressed, noisy) so the model sees variety.
calibration	Making the 0-1 score mean a real probability, so thresholds are meaningful.
domain shift	Test audio looks different from training audio. THIS is our core problem.